

<https://github.com/NAVYA6565/DBMS-.git>

Subquery 1 — Students scoring above the Average

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks > (SELECT AVG(Marks) FROM Student);
```

Result Grid					
	Student_ID	Name	Dept_ID	Marks	City
▶	102	Neha	20	92	Mumbai
	104	Manya	30	88	Bangalore
	105	Tanisha	20	95	Delhi
✱	NULL	NULL	NULL	NULL	NULL

Student 1 x					
Output					
Action Output					
	#	Time	Action		
✓	6	20:02:33	CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(50), ...		
✓	7	20:02:33	INSERT INTO Department (Dept_ID, Dept_Name) VALUES (10, 'CSE'), (20, 'ECE'),...		
✓	8	20:02:33	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...		
✓	9	20:02:33	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...		

Outer:

```
CREATE DATABASE IF NOT EXISTS college_dbms;
```

```
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
```

```
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
```

```
Dept_ID INT PRIMARY KEY,
```

```
Dept_Name VARCHAR(50)
```

```
);
```

```
CREATE TABLE Student (
```

```
Student_ID INT PRIMARY KEY,
```

```
Name VARCHAR(50),
```

```
Dept_ID INT,
```

```
Marks INT,
```

```
City VARCHAR(50)
```

```
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
```

```
(10, 'CSE'),
```

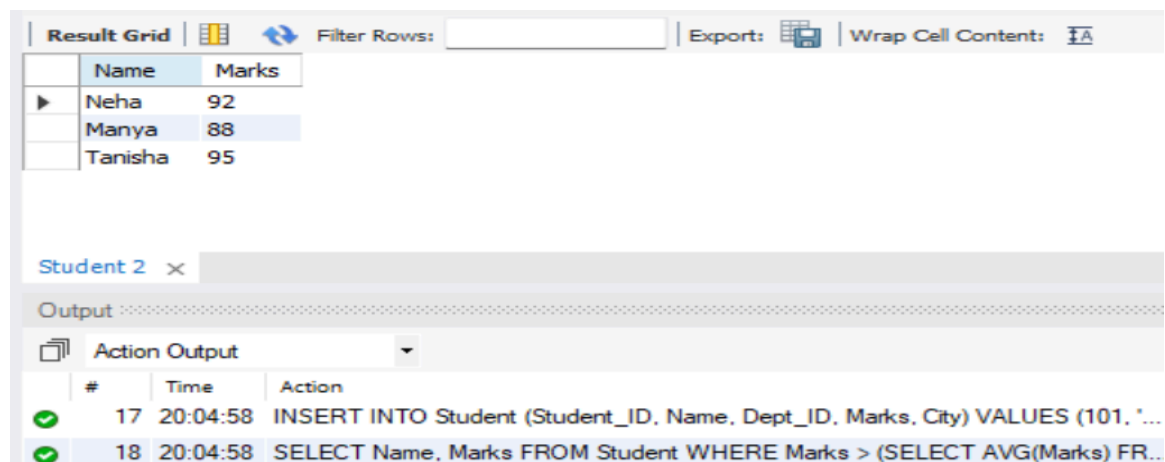
```
(20, 'ECE'),
```

```
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
```

```
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Name, Marks FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student);
```



The screenshot shows a database management interface. At the top, there's a 'Result Grid' tab with a table containing three rows of student data. Below this, there's a tab labeled 'Student 2'. Underneath, there's an 'Output' section with a dropdown menu set to 'Action Output'. This section displays a log of two actions: an INSERT statement (Action #17) and a SELECT statement (Action #18) that matches the SQL code provided in the previous blocks.

	Name	Marks
▶	Neha	92
	Manya	88
	Tanisha	95

#	Time	Action
✓ 17	20:04:58	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...
✓ 18	20:04:58	SELECT Name, Marks FROM Student WHERE Marks > (SELECT AVG(Marks) FR...

Inner:

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
```

```
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Name, Marks FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student);
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	Name	Marks
▶	Neha	92
	Manya	88
	Tanisha	95

Student 3 ×

Output

Action Output ▼

	#	Time	Action
✓	26	20:06:10	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...
✓	27	20:06:10	SELECT Name, Marks FROM Student WHERE Marks > (SELECT AVG(Marks) FR...

Subquery 2 — Student(s) who scored the highest marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks = (SELECT MAX(Marks) FROM Student);
```

Result Grid		Filter Rows:	Edit:   			Export/Import: 	
	Student_ID	Name	Dept_ID	Marks	City		
▶	105	Tanisha	20	95	Delhi		
*	NULL	NULL	NULL	NULL	NULL		

Student 4 x

Output

Action Output

#	Time	Action
✓ 35	20:08:57	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...
✓ 36	20:08:57	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...

Inner:

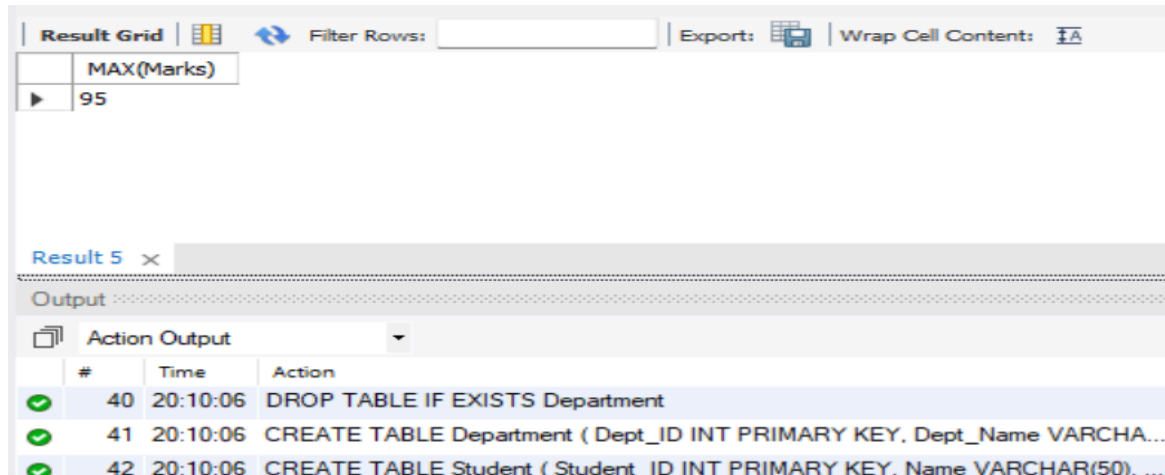
```

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),

```

(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT MAX(Marks) FROM Student



The screenshot shows a database management interface. At the top, there's a toolbar with 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. Below this, a table displays the result of a query: 'MAX(Marks)' with a value of '95'. Below the table, there's a section labeled 'Result 5' with a tab 'Output'. Under 'Output', there's a dropdown menu set to 'Action Output'. Below this, a table shows the execution log with columns '#', 'Time', and 'Action'. The log contains three entries, all marked with a green checkmark, indicating successful execution.

#	Time	Action
40	20:10:06	DROP TABLE IF EXISTS Department
41	20:10:06	CREATE TABLE Department (Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(50))
42	20:10:06	CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(50), Marks INT, City VARCHAR(50))

Subquery 3 — Students scoring above Rahul

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
```

```

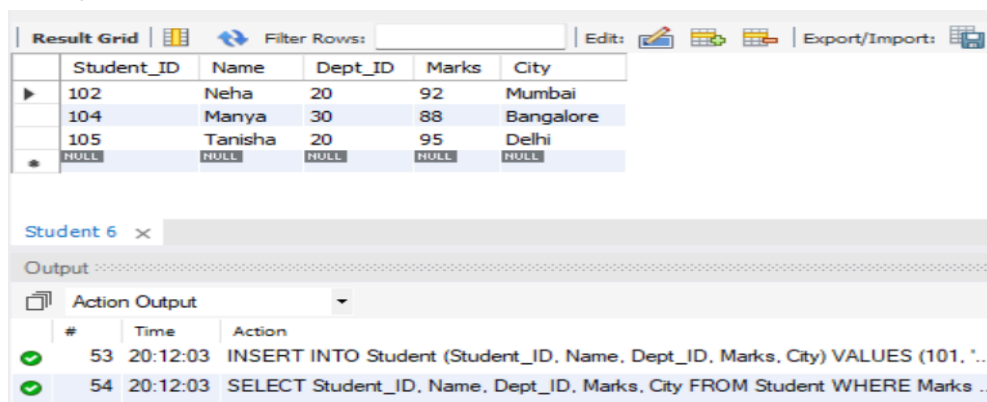
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

```

```

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT Marks FROM Student WHERE Name = 'Rahul' OR Name =
'Navya');

```



The screenshot shows a database management interface. At the top, there's a 'Result Grid' with a table containing 5 columns: Student_ID, Name, Dept_ID, Marks, and City. The table has 5 rows of data and one row of NULL values. Below the result grid, there's an 'Output' section with a tab labeled 'Action Output'. This section shows a log of database actions, including an INSERT statement (Action 53) and a SELECT statement (Action 54) that matches the SQL code provided in the previous blocks.

Student_ID	Name	Dept_ID	Marks	City
102	Neha	20	92	Mumbai
104	Manya	30	88	Bangalore
105	Tanisha	20	95	Delhi
NULL	NULL	NULL	NULL	NULL

#	Time	Action
53	20:12:03	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...
54	20:12:03	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...

Subquery 4 — Students belonging to the same department as Rahul

```

CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)

```

```

);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Dept_ID =
(SELECT Dept_ID FROM Student WHERE Name = 'Navya');

```

Result Grid						Filter Rows:		Edit:		Export/Import:	
	Student_ID	Name	Dept_ID	Marks	City						
▶	101	Navya	10	85	Delhi						
	103	Spoorthi	10	76	Delhi						
*	NULL	NULL	NULL	NULL	NULL						

Student 7 ×

Output

Action Output

#	Time	Action
62	20:13:12	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...
63	20:13:12	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Dept_I...

Subquery 5 — Students who scored above the average CSE marks

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
  Dept_ID INT PRIMARY KEY,
  Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
  Student_ID INT PRIMARY KEY,
  Name VARCHAR(50),
  Dept_ID INT,
  Marks INT,
  City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks >
(SELECT AVG(Marks) FROM Student WHERE Dept_ID = 10);
```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
Student_ID	Name	Dept_ID	Marks	City	
101	Navya	10	85	Delhi	
102	Neha	20	92	Mumbai	
104	Manya	30	88	Bangalore	
105	Tanisha	20	95	Delhi	
NULL	NULL	NULL	NULL	NULL	

Student 8					
Output					
Action Output					
#	Time	Action			
71	20:14:18	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...			
72	20:14:18	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...			

Subquery 6 — Students who belong to a department having a student from Mumbai

```
CREATE DATABASE IF NOT EXISTS college_dbms;
```

```
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
```

```
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
```

```
Dept_ID INT PRIMARY KEY,
```

```
Dept_Name VARCHAR(50)
```

```
);
```

```
CREATE TABLE Student (
```

```
Student_ID INT PRIMARY KEY,
```

```
Name VARCHAR(50),
```

```
Dept_ID INT,
```

```
Marks INT,
```

```
City VARCHAR(50)
```

```
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
```

```
(10, 'CSE'),
```

```
(20, 'ECE'),
```

```
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
```

```
(101, 'Navya', 10, 85, 'Delhi'),
```

```
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Dept_ID IN
(SELECT Dept_ID FROM Student WHERE City = 'Mumbai');
```

Result Grid	Filter Rows:	Edit:	Export/Import:		
	Student_ID	Name	Dept_ID	Marks	City
▶	102	Neha	20	92	Mumbai
	105	Tanisha	20	95	Delhi
✱	NULL	NULL	NULL	NULL	NULL

Student 9 x

Output

Action Output

#	Time	Action
80	20:15:54	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...
81	20:15:54	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Dept_ID...

Subquery 7 — Students who scored higher than every ECE student

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (
Dept_ID INT PRIMARY KEY,
Dept_Name VARCHAR(50)
);
CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
```

```

Dept_ID INT,
Marks INT,
City VARCHAR(50)
);
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
(30, 'MECH');
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Navya', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Spoorthi', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');

```

```

SELECT Student_ID, Name, Dept_ID, Marks, City
FROM Student
WHERE Marks > (
SELECT MAX(Marks)
FROM Student
WHERE Dept_ID = 20
);

```

Result Grid					
Filter Rows:					
Edit: Export/Import:					
	Student_ID	Name	Dept_ID	Marks	City
*	NULL	NULL	NULL	NULL	NULL

Student 10 x					
Output					
Action Output					
	#	Time	Action		
✓	89	20:16:50	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, '...		
✓	90	20:16:50	SELECT Student_ID, Name, Dept_ID, Marks, City FROM Student WHERE Marks ...		

